Master thesis

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A Three-Pipe 4GDHC System

Simultaneous Heating and Cooling with Waste Heat Recovery

Thesis in Construction Engineering with Specialisation in Renewable Energy, 30 Credits

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Summary

Cities that need both heating in winter and cooling in summer, often at the same time, usually build four separate pipes to carry hot and cold water. This thesis studies an alternative with three pipes: one carrying hot water at 60ºC, one carrying chilled water at 6ºC and one shared return pipe. External heat is essential, since a heat pump only moves heat and cannot create it. At the central plant, a heat pump uses the water coming back in the shared return pipe together with waste heat from the data centre to produce both the hot and cold supply. The calculations show that the heat pump works efficiently, delivering roughly four units of heat for each unit of electricity. The main drawback is some heat leaking from the hot pipe to the cold pipe through the soil. The proposed three-pipe layout costs about 39% less per metre to install than four pipes. It is a promising option for mixed-use districts where waste heat is available, but it needs testing in a pilot installation.

Keywords: District heating and cooling, three-pipe system, heat pump coefficient of performance (COP), Lorenz cycle mixed return temperature, fourth-generation district heating, inter-pipe heat transfer

Abstract

Research on Fourth-generation district heating and cooling (4GDHC) has largely focused on low-temperature single-service networks or conventional four-pipe configurations, leaving simultaneous heating and cooling delivery through shared infrastructure unexplored. This thesis defines and analytically evaluates a unidirectional three-pipe system in which a heating supply pipe (60ºC), a shared common return pipe, and a cooling supply pipe (6ºC) are served by a centralised dual-evaporator heat pump, with a Y-separator dividing the mixed return between the evaporator and condenser sides, supplemented by data centre waste heat provides the external heat input that the heat pump requires. This study comprises an analytical model with multipipe thermal resistance and superposition heat loss model, a mass-flow-weighted mixed return temperature model, and Carnot and Lorenz cycle coefficient of performance (COP) methods were used to quantify distribution heat loss, inter-pipe interference along the pipe temperature drop, heat pump performance across design off-load scenarios, and thermal storage requirements. The system was benchmarked against a conventional four-pipe network and the Ranagård three-pipe installation using a five-category installed cost model. Results indicate a stable heat pump coefficient of performance (COP) of approximately 3.0 – 4.0. The cold pipe gained about 129 W per meter, almost entirely from the adjacent hot pipe, a penalty that greater spacing can reduce. Installed cost was about 39% lower than the four-pipe alternative.

Preface

This thesis marks the end of a long and meaningful journey; one we did not walk alone.
We want to thank God first, for getting us through the harder stretches of this process. Thanks to our families for putting up with our stress, our late nights and probably a lot of complaints and for supporting us anyway.
We also want to thank our supervisor, Helge Averfalk. The idea for this thesis came from their earlier work, and they were generous enough to let us take it further. Thanks also for answering our emails fast and even telling us straight when something in the thesis wasn't working.
We wrote this thesis together, as friends, and we are glad we had each other through it.

Halmstad, September 2026
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1 Introduction

1.1 Background and motivation

District heating systems are an important part of sustainable energy systems, especially in regions with cold climates. Over the years, district heating has developed into different generations, where fourth-generation district heating (4GDH) focuses on lower supply temperatures, higher energy efficiency, and better integration of renewable energy and waste heat sources. Lower temperature operation helps in reducing heat losses in the network and improves the performance of heat pumps. With the proposition of energy conservation and environmental protection policies in various countries, smart energy systems and fourth-generation district heating and cooling technologies have gradually become hot research topics (Xu, et al., 2021). As cities grow denser and the need for both space heating in winter and space cooling in summer intensifies, conventional district energy infrastructure faces significant technical and economic challenges.

District heating (DH) systems have evolved through successive generations since their inception in the late 19th century. First-generation systems relied on steam distribution, while second- and third-generation networks progressively adopted lower supply temperatures and improved insulation technologies. The fourth generation of district heating (4GDH), as conceptualised by (Lund, et al., 2014), represents a paradigm shift: it operates at low supply temperatures (below 70°C), integrates renewable energy sources, and is designed to be compatible with low-energy buildings and waste heat recovery. (Lund, et al., 2014) emphasise that the development of future district heating systems and technologies involves energy savings and conservation measures as an important part of the technology.

However, a critical challenge arises in urban districts where simultaneous heating and cooling loads coexist, such as commercial offices, residential apartments and retail spaces. A traditional approach to serve this area typically laying parallel networks. A 4-pipe system comprises of two pipes for heat supply and heat return, and two 4-pipe for cold supply and cold return, together referred as a 4-pipe system. But this configuration involves high infrastructure investment and complex substation design, and these limitations force for exploring alternative distribution technologies. One such alternate is three-pipe district heating system introduced in (Averfalk & Werner, 2018). In this configuration, a single shared supply pipe replaces the two heating supply pipes, while the two return pipes remain separate. When this three-pipe principle is instead applied to combines heating and cooling networks, the configuration can be reversed, two supply pipes serves the hot and cold demand separately, while a shared common return pipe. The theoretical approach lies in the thermal behaviour of heating and cooling consumers, the hot return from cooling consumers and cold return from heating consumers can mix to form a low return temperature which will benefit the heat pump performance at the plant.

Although, this concept has been explored in limited number of studies, both analytical and computational analyses. Previous research primarily focuses on district heating alone, using one such configuration with hot supply and return, rather than considering both heating and cooling simultaneously. Here, in this thesis, both heating and cooling are analysed simultaneously on the same system, also comparing over the existing conventional district heating systems. Based on three-pipe systems, this paper focuses on studying dual supply of both heating and cooling, with a common return.

1.2 Problem statement

Existing fourth-generation district heating and cooling (4GDHC) literature primarily focuses on low-temperature heating networks, or conventional 4-pipe configurations. The three-pipe unidirectional system, in which Pipe 1 supplies heating (~60°C), Pipe 2 serves as common return (~20°C), and Pipe 3 supplies cooling (~6°C), takes a design space that combines the reduced infrastructure when comparing with traditional district heating and cooling network. Despite its theoretical advantages, this system configuration presents several unresolved technical challenges. The mixing of warm heating and cooling return water in single pipe creates a variable return temperature profile that depends on the simultaneous heating and cooling loads, the pipe geometry, pipe layout and thermal characteristics of the building substation. Poor management of the mixed return temperature can badly affect heat pump coefficient of performance coefficient of performance (COP) and system thermal efficiency. Additionally, distance between the heating supply pipe and the cooling supply pipe introduces the risk of unwanted inter-pipe heat transfer, which badly affect cooling quality and overall system performance.
A thorough theoretical and comparative analysis of this system is thus required, which helps to quantify the temperature differential (ΔT) improvements, heat pump efficiency gains, heat recovery potential, and pipe heat transfer losses under different load scenarios

1.3 Research questions

This thesis is structured around the following primary question:

Can a three-pipe district heating and cooling configuration improve temperature differences and system efficiency compared with conventional parallel networks?

To address this question comprehensively, the following sub-questions help the analytical and comparative work of this thesis.

1. What are the potential advantages and practical limitations of a three-pipe system?
2. How does the shared return pipe affect the mixed return temperature and the system wide temperature differential (ΔT) under different load scenarios?
3. What is the theoretical potential for heat recovery between heating and cooling consumers through shared return loop?
4. How does pipe heat transfer influence system performance?
5. How does the proposed three-pipe configuration compare in efficiency, cost and technical complexity with existing conventional four-pipe parallel network and the Ranagård three-pipe installation?

1.4 Research objectives

In response to the research questions stated above, this thesis pursues the following objectives.

1. To define and characterise the proposed three-pipe district heating and cooling system in terms of physical layout, operating principles and design assumptions.
2. To conduct a systematic literature review of existing three-pipe and fourth generation district heating system, including Ranagård three-pipe installation.
3. To develop a steady-state analytical model for evaluating system temperature differentials, heat pump coefficient of performance (COP), pipe heat transfer losses and mixed return temperatures under multiple load scenarios.
4. To compare the proposed system against conventional 4-pipe system, and Ranagård configurations using system efficiency, installation and maintenance cost, and

operational complexity.

5. To identify the technical conditions under which the proposed three-pipe offers a competitive alternate to existing district heating and cooling systems.

1.5 Scope and delimitations

This thesis focuses on the theoretical analysis and comparative evaluation of the three-pipe district heating and cooling system. The analysis is based on steady-state energy balance modelling and does not include dynamic transient simulation or real-time operational data. Assumed input data including supply temperatures, flow rates, building load profiles and pipe thermal properties, which are derived from published literature and standard engineering references where field measurements are unavailable.

The study is delimited to represent an urban district with simultaneous heating and cooling demand. It does not address the detailed structural design of pipe network layouts, geotechnical installation conditions or regulatory frameworks specific to single country. Economic analysis is treated at high level, focusing on relative capital cost comparisons rather than lifecycle cost assessments. While the Ranagård installation serves as reference benchmarks, this thesis does not have access to proprietary operational data from these systems. Comparisons are therefore based on publicly available performance data and published studies.

1.6 Significance of study thesis structure

This thesis is organised into six chapters. Chapter 1 (the present chapter) introduces the research context, problem statement, research questions, objectives, and scope. Chapter 2 presents a comprehensive review of the literature of fourth-generation district heating, three-pipe system concepts, heat pump integration, and existing reference systems including the Ranagård installation. Chapter 3 provides a detailed definition of the proposed three-pipe system, including physical layout, pipe temperature specifications, design assumptions and operating modes.

Chapter 4 describes methodology used, including the energy balance equations imparting return temperature mixing and heat pump performance, and the scenarios used for comparative analysis. Chapter 5 presents the results of the analysis, covering temperature differential performance, estimated heat pump coefficient of performance (COP), pipe heat transfer losses, and heat recovery potential across the defined load scenarios. Chapter 6 comprises all the findings, addresses each research question, discusses limitations and proposes insights for future research.

2 Literature Review

2.1 Introduction to literature review

This chapter presents a review of the existing knowledge relevant to the proposed three-pipe fourth-generation district heating and cooling (4GDHC) system. The review begins from foundational concepts and advances toward the technical gaps addressed by this thesis. This thesis begins with the evolution of district heating systems and the principles of fourth-generation district heating, then forward to multi-pipe configurations including heat loss modelling of twin-pipe and three-pipe systems, and finally heat pump integrations in district heating networks. The chapter then reviews one reference system, the Ranagård installation, before concluding with the research gaps that motivate the study.

2.2 Evolution of district heating systems: From first to fourth generation

The concept of the district heating dates to the late 19th century, when steam-based systems appeared in New York and parts of Northern Europe. First Generation (1GDH) systems distributed steam at very high temperatures, subjected to safety risks and heat losses. Second generation system (2GDH) adapts pressurised hot water at temperatures above 100°C. This generation (3GDH) systems dominant from the 1970s onward proposed pre-insulated pipes and supply temperature in range of 70-90°C.
Fourth-generation district heating (4GDH) is highlighted by low supply temperature (below 70°C), easiness with heat pumps, integration with smart energy systems and ability to back low temperature renewable and waste heat (Lund, et al., 2014).

2.3 Multi-pipe district heating configurations and heat-loss modelling

2.3.1 Conventional two-pipe and four-pipe systems

The two-pipe system (one supply, one return), is the most common DH concept, but cannot serve both district heating and district cooling simultaneously. The conventional idea is a four-pipe network with fully separate heating and cooling pipes, avoiding thermal interference between pipes but requires more excavation (Lund, et al., 2014).

2.3.2 Heat-loss modelling of twin-pipe and pre-insulated single pipes

A practical gap in this literature is that most heat loss studies focus on analytical or numerical models, and few are validated against laboratory measurements relevant to real networks. (Jakubek, et al., 2023) built an experimental test rig comparing twin-pipe and single pre-insulated pipe configurations at diameters of DN40, DN50 and DN65. By supplying water at 90°C and returning at 50°C, their study compared measured heat losses between a 1D analytical model based on linear thermal resistance, and for twin pipes a mutual influence resistance between supply and return pipe.
The findings are relevant to this thesis. First, the analytical model agreed with the laboratory measurements within about 5-10% of all pipe sizes and configurations, confirming that simple 1D thermal resistance models remain usable for practical network design even at larger diameters. Secondly, twin-pipe configurations consistently showed a lower total heat loss than two separate single pre-insulated pipes of the same diameter, because of the shared insulation and mutual thermal resistance between supply and return pipe.
The authors also carried out an economic comparison for a real case study, a reheat system in

Krakow, which links an underground thermal storage tank to a ground-source heat pump. Even higher, the twin-pipe was more expensive option to install, its lower heat loss produced more in energy savings, giving a payback of about five years and a positive Net Present Value (NPV) after fifteen years. This backs the general principle applicable throughout in this thesis, that a pipe configuration with lower heat loss can still be the economically viable option once operating savings outweigh the higher installation cost.

2.3.3 Heat-loss modelling of two-supply, one-return three-pipe systems

Two papers by Xu and co-authors develop heat-loss models specifically for the “two-supply, one-return" arrangement that is closest to the system proposed in this thesis, although in both cases the two supply pipes carry water at different heating temperatures rather than one heating and one cooling system. (Xu, et al., 2021) proposed the TOTS (two-supply/one return, triple-pipe) arrangement, in which a domestic hot water pipe and a space heating pipe are shared together in one insulation, and the return pipe separately. Due to the unavailability of an existing method, they first derived and validated separate heat loss models for (1) two parallel single pipes buried side by side (multipole method) and (2) two pipes sharing one insulation. Then combined these solutions, considering the paired supply pipes as a single equivalent pipe of same outer size. They also built a 3D finite element numerical model. Both the analytical and numerical solutions agreed with the reference heat loss data to give confidence in the modelling approach.

The main design finding from this paper concerns pipe spacing. As the distance between the two-supply pipe increases, the total heat loss of the system rises, because the pipes are closer to the outer edge of insulation near to surrounding soil. But the heat exchange between the two supply pipes falls because of the increased thermal resistance between them. In their study, reducing the space from 114.1 mm to 84.1 mm cut total heat loss from 24.1 W/m to 20.2 W/m but almost doubled the mutual heat exchange (from 0.82 W/m to 1.52 W/m). The authors conclude that pipe spacing cannot be chosen to minimise the heat loss alone, a tightly spaced pair of supply pipes loses less overall heat but transfers more heat between the pipes themselves. This is directly relevant to this thesis that the hot supply pipe could raise the temperature of an adjacent cold supply pipe. They also found that the return pipes heat loss was almost unaffected by supply-pipe spacing since it is buried separately.

A follow-up study in (Xu, et al., 2026), examined the same three-pipe layout used terms HTP= high temperature supply pipe, LTP= low temperature supply pipe and RP= return pipe. Apart from heat-loss studies, this paper follows combined thermodynamic- economic parametric study, using utilization efficiency about the heat usefully delivered to and returned from the customer and the heat lost in transmission and inter-pipe exchange. Using a validated CFD model, they varied the temperature difference between the HTP and LTP and the HTP flow velocity independently and mutually.

Two findings for this thesis:

1. Temperature difference dominates thermal performance; flow velocity has less impact. By increasing HTP and LTP temperature differences from 0 to 40°C increased total system heat loss from 2.34 W/m to 2.53 W/m and also increased inter-pipe heat exchange from 0 W/m to 0.77 W/m. Constantly, changing the HTP flow velocity across 0.25-1.25 m/s range left total heat loss and inter pipe heat exchange unchanged noticeably. This means that, from a thermal point of view, minimizing the difference the heating and cooling supply temperatures is the key factor for controlling the inter pipe exchange.
2. The lowest heating cost occurred at a large temperature difference combined with highest flow velocity. This gives a minimal annual cost of about 4,970 USD /year in their case study. Because at high flow velocity, pumping cost savings outweigh the

extra heat cost. This is a useful aspect for Chapter 5 of this thesis, this shows that lowest heat loss and lowest operating cost do not occur at the same point, and any study on three-pipe systems should consider both metrics rather focusing on heat loss alone.

Together, these (Xu, et al., 2021) (Xu, et al., 2026) papers give this thesis a validated modelling approach for a three-pipe layout, adding quantitative evidence. Also, showing how supply pipe spacing and the temperature difference between adjacent supply pipes factors inter-pipe heat exchange. Since their configuration uses two heating supply temperatures rather than one heating and one cooling supply temperatures, the magnitude of inter-pipe heat transfer they validated cannot be assumed directly to this thesis three-pipe system. The temperature differential in the proposed three-pipe system in this thesis has roughly 60°C hot supply to 6°C cold supply, considerably larger than 0-40°C s range mentioned in (Xu, et al., 2021).

2.4 Return temperature management and temperature differential performance

Return temperature is one of the most crucial operational parameters in a district heating, a lower return temperature increases the achievable ΔT, eventually reducing pumping energy and mass flow leading to improved efficiency of both CHP and heat pump. In a three-pipe system with a shared return, the return temperature reflects a weighted mix of warmer return water from cooling consumers and cooler return water from heating consumers, varying on seasonal balance of heating and cooling demand.

2.5 Heat pump integration in district heating networks

2.5.1 Heat pump performance fundamentals and the role of temperature lift

A study by (Schlosser, et al., 2020) reviewed 155 documented case studies and 33 market available large scale heat pump products to characterise how heat pump performance coefficient of performance (COP) depends on the temperature lift (ΔT_{lift}) between the heat source and heat sink and to develop economic feasibility for the heat pump integration. This directly supports the central argument in this thesis that a moderate return temperature that can be provided by a shared return three-pipe system improves the heat pump coefficient of performance (COP) relative to a colder source.

The paper criticises the commonly used approach of estimating coefficient of performance (COP) from the Carnot cycle using second-law efficiency, assumed at around 0.45. Across their database from 33 heat pumps, this constant efficiency approach performed badly, especially at the low and high ends of temperature lift. So, an empirical model was fitted for three heat pump categories.

1. Standard/high temperature heat pumps (SHP/HTHP) with synthetic organic refrigerants- valid for$10\ K \leq \Delta T_{lift} \leq 78K$, sink temperature up to 100°C
2. Very high temperature heat pumps (VHTHP) - valid for $25\ \text{K} < \Delta T_{lift} < 95\ \text{K}$, sink temperatures up to 160°C
3. Ammonia (R717) heat pumps – valid for $30K \leq \Delta T_{lift} \leq 75k$, sink temperatures 70-85°C

They developed a similar model for CO_2 heat pumps based on logarithmic mean sink/source temperatures, since the refrigerant temperature in a CO_2 gas cooler is not constant.

Across all the reviewed case studies, coefficient of performance (COP) values in the 3-6 range were the most favourable payback period (typically 3-5 years), generally featuring sink temperatures below 100°C, temperature lift below 50 K and more than 6000 operating hours per year.

2.5.2 Economic feasibility and break-even analysis

The cost of heat from a heat pump against a standard gas boiler as a function of temperature lift and ration of electricity to natural gas price was compared in (Schlosser, et al., 2020) . For a European average electricity/gas price ratio of about 3.5, they found economic break-even at a temperature lift of roughly 34-39 K, but for more favourable price ratios break-even shifted to a 43-50 K lift. The paper argues that feasibility studies should use empirically fit coefficient of performance (COP) like theirs rather than a constant efficiency assumption. A methodological approach this thesis should follow in Chapter 4 when estimating the coefficient of performance (COP) of the heat pump.
The paper also confirms the thermodynamic mechanism this thesis relies on the heat recovery concept of the heat pump. Because coefficient of performance (COP) falls as temperature lift increases, a heat pump drawing from a shared return requires a smaller lift to reach the heating supply temperature compared to colder air or ground source. Therefore, achieve a higher coefficient of performance (COP), as mentioned in the paper, where systems achieving lower return/source temperatures below 40°C consistently produced higher coefficient of performance (COP).

2.6 The Ranagård Three-pipe installation: A Case study

The Ranagård installation in Sweden is one of the few documented real-world three-pipe networks, through its one supply pipe and two return pipes carry water at different heating temperatures rather than one heating and one cooling system- the same structural difference noted above for (Xu, et al., 2021) (Xu, et al., 2026) three-pipe studies. Since Ranagård uses separated rather than mixed return flows, this does not directly validate the shared return configurations studied in the thesis. However, provide real-world evidence that a three-pipe network can achieve very low return temperature, and it may also offer potential capital cost savings compared with conventional systems, although these savings require further investigation.

2.7 Identification of research gaps

The literature reviewed in this chapter confirms and acknowledges three gaps that motivate this thesis:

1. No published study examines a three-pipe configuration combining one heating supply pipe, one cooling pipe, and a shared return in a 4GDHC context. (Xu, et al., 2021) use a configuration with two heating supply pipes and one common return, while the Ranagård case uses the opposite configuration, one heating supply pipe and two separate return pipes. In both cases only heating not combined heating and cooling is addressed. (Xu, et al., 2021) validated multipole/thermal resistance modelling method is directly usable in this thesis, but their temperature differential, validated against 0-40°C is smaller to the proposed system. This thesis use of their modelling approach at a larger temperature differential than they validated should not be a direct extension.
2. No analytical model exists to quantify how much cooling rejection can be usefully recovered via a heat pump operating on a mixed return flow.
3. No systematic comparison exists positioning a heating-cooling three-pipe configuration against the full range of alternative ideas- four-pipe and Ranagård focusing on heat loss, inter-pipe heat exchange, heat pump coefficient of performance (COP) and cost.

3 System Definition and Design

3.1 Introduction

This chapter provides a detailed technical definition of the proposed three-pipe fourth-generation district heating and cooling (4GDHC) system. Building on the theoretical concept established in Chapter 2 Literature Review, the physical layout, pipe roles and operating temperatures, flow configurations, substation connection and design assumptions leading to the analytical model developed in Chapter 4. Also, the chapter analyses the proposed design against the conventional four-pipe system to highlight the structural differences that motivate the comparative analysis in Chapter 6.

The proposed system is characterised as a unidirectional three-pipe network in which thermal energy for both heating and cooling is distributed from a central energy plant to building substations through three buried pipes: a heating supply pipe (Pipe 1), a shared common return pipe (Pipe 2), and a cooling supply pipe (Pipe 3). This new configuration eliminates one pipe compared with the conventional four-pipe system, with the shared return receiving mixed return flows from both heating and cooling customers simultaneously. The system is designed in the way that mixed return pipe (Pipe 2) is not passed directly to central plant, instead it is divided by a Y-separator and routed through the central heat pump which produces hot supply for pipe 1 and cold supply for pipe 3. Since a heat pump does not produce heat, an external energy input is required. The compressor is driven by electricity, the heat for the evaporator comes from the cooling return and the waste heat recovered from the data centre.

The central heat pump draws on two independent heat sources rather than one. The main source is waste heat recovered from data centre, supplied to the plant at approximately 25°C and the other source is the Pipe 2 return mix at approximately 18-23°C. The two sources are connected to two separate evaporators circuits of a dual-evaporator heat pump, with the refrigerant sides of both evaporator circuits joined at a common compressor, so that the two heat streams are combined within the machine rather than mixed on the water side. This dual source arrangement increases the total heat available for heating and cooling demand, also provides a comparatively stable warm heat source.

3.2 Conceptual overview of the proposed System

3.2.1 Design philosophy

The design of the proposed three-pipe system is guided by three engineering objectives. The first objective is the capital cost reduction: by replacing the two separate return pipes of a conventional four-pipe system with a single shared return pipe, the total pipe count is reduced from four to three, helps savings in material, trench excavation, insulation and installation labour. For a district of comparative urban population, this reduction represents a noticeable share of the total investment cost.

The second objective is thermal efficiency improvement through higher temperature differential (ΔT). In a conventional four-pipe system, the heating return temperature and the cooling return temperature are entirely independent and cannot benefit from each other. In the proposed system, the mixing of the heating and cooling return in a single pipe produces a moderate mixed return temperature (18-23°C). The mixed return water of the network is a source to central heat pump. A Y separator splits the flow to the evaporator and condenser side. In the evaporator, heat is absorbed from the water, which leaves it colder and is supplied to the cooling distribution. In the condenser, heat is added to the water, which leaves warmer and supplies the heating distribution. The energy delivered to the customers is supplied by the

electricity used by the heat pump and by the waste heat from the data centre, which is added into the network. This improves the coefficient of performance coefficient of performance (COP) of the heat pump relative to systems that use ambient air or ground as the heat source or sink.
The third objective is heat recovery integration. The shared return pipe creates a continuous thermal bridge between the cooling and heating stream. Under simultaneous heating and cooling load during shoulder seasons, a portion of the heat rejected from cooling consumers can be directly recovered and supplied to heating consumes, reducing the primary energy required from the central plant. This internal heat recovery mechanism is a main thermodynamic advantage of the three-pipe idea over conventional four-pipe system.

3.3 System boundary and scope

The system boundary for this study includes the primary distribution network from the central energy plant to the building level heat exchanger substations, inclusive of two distribution pipes, one shared return pipe, the central plant dual evaporator heat pump, a Y-separator, the interface point at which the data centre waste heat loop connect s to evaporator unit, a storage tank and the substation interface. It does not extend to the secondary distribution within the individual buildings nor to the primary energy input outside the central plant.
In this thesis, a data centre is used as an example of industrial waste heat recovery. Industries often produce heat that is lost to the surroundings. A data centre is a good example because almost all electricity used by the servers becomes heat, and this heat is produced throughout the year. The heat pump takes heat from two sources. One is mixed return from the shared return pipe, where heat is absorbed from the cooling-demanded customers and the other is the waste heat from the data centre cooling loop at about 25°C, which is supplied to a separate data centre side evaporator of the heat pump. The data centre does not add heat to the network pipes directly. Its heat enters the system through the heat pump.
Since the return flow has different mass flow depending on demand scenarios, the mass flow required for the heat pump might be less than the mass flow in the return flow. So, a storage tank before the Y-separator to store the excess water with the mass flow which can be used when required.
The network is modelled as a single distribution segment of representative length, for a mixed residential and commercial district with calculated peak heating and cooling demand profiles.

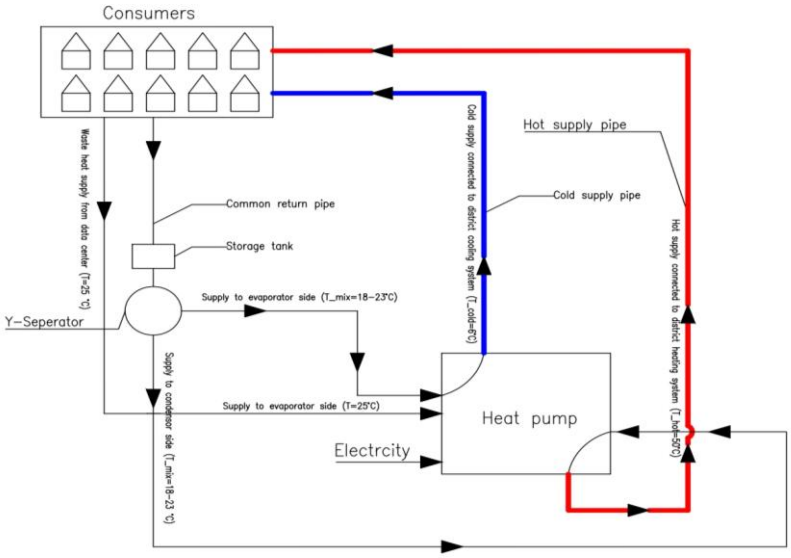


Figure 1: System definition.

3.4 Pipe configuration and operating temperature

3.4.1 Pipe 1: Heating supply pipe

Pipe 1 is the heating supply pipe, carrying pressurised hot water from the central energy plant to building substations with a space heating or domestic hot water (DHW) demand. In line with fourth-generation district heating ideas, the design supply temperature for Pipe 1 is set at 60°C. This temperature is sufficient to serve space heating systems operating in low-temperature scenarios, and it is consistent with the supply temperature range recommended by (Lund, et al., 2014) for fourth- generation networks.

The choice of 60°C as the heating supply temperature reflects a deliberate design trade-off. A higher supply temperature would increase the heating capacity per unit mass flow and reduces the interface with existing building heating systems, but this might also increase heat losses from the pipe and leads to reduced temperature difference between Pipe 1 and Pipe 3. A lower supply temperature would improve the coefficient of performance (COP) of the heat pump and reduce distribution losses but may be insufficient to meet domestic hot water demand (typically 45°C) without additional boosting. The 60°C set point therefore represents an optimised temperature for the mixed urban use assumed in this thesis.

3.4.2 Pipe 2: Return pipe

Pipe 2 is the common return pipe, receiving mixed return flows simultaneously from heating customers (Pipe 1 circuit) and the cooling customers (Pipe 3). It is the defining structural feature of the proposed system and the principal source of its thermodynamic advantages over the four-pipe configuration. The temperature of the fluid in Pipe 2 at any given point in the network is determined by the mixing of the heating return flow (approximately 30-35°C) and the cooling return flow (approximately 12-16°C). The mixed temperature depends on the relative mass flow rated of the two return streams and varies with the instantaneous ratio of heating to cooling load across the served area. Under peak winter conditions, the heating return flow dominates, and the mixed return temperature approaches the upper end of the range 23°C. Similarly, under peak summer conditions, the cooling return flow dominates, and the mixed temperature approaches the lower end of 18°C. During shoulder seasons, the two streams mix to produce an intermediate temperature that is particularly good for central heat pump operations. The dynamic behaviour of Pipe 2 is modelled analytically in Chapter 4 using mass flow weighted equations.

3.4.3 Pipe 3: Cooling supply pipe

Pipe 3 is the cooling supply pipe, distributing chilled water from the central cooling plant to building substations with a space cooling or process cooling demand. The design supply temperature for Pipe 3 is set at 6°C, which is standard for the district cooling applications and compatible with commercial buildings. The cooling supply flow returns to Pipe 2 at a design return temperature of approximately 18-23°C, representing a temperature rise of 12-17°C across the cooling substation. This temperature differential is established with district cooling practice and reflects good substation design with appropriate heat exchanger sizing. The warm return from Pipe 3 customers is the primary contributor to the elevated component of the mixed return temperature in Pipe 2.

A critical design consideration for Pipe 3 is its proximity to Pipe 1 within the buried pipe bundle. The temperature difference between the heating supply (60°C) and the cooling supply (6°C) is around 54°C, creating a substantial driving force for inter-pipe heat transfer through the soil. This heat transfer, not properly controlled through insulation and pipe separation, may cause the cooling supply temperature to rise along the pipe route, degrading the cooling

capacity delivered to the substations and increasing the energy consumption at the central plant. The mitigation strategies for this situation are discussed in the section Inter-Pipe heat transfer and Mitigation of the same chapter.

Table 1: Design parameters for each pie in the proposed three-pipe system.

Parameter	Pipe 1 (Hot supply)	Pipe 2 (Shared return)	Pipe 3 (Cold supply)
Fluid temperature (Supply	60°C	N/A	6°C
Design return temperature	30-35°C (at customer)	18-23°C (mixed in network)	12-16°C (at customer substation)
Flow direction	Central plant to customers	Customers to central plant	Central plant to customers
Pipe nominal diameter(assumed)	DN65	DN65	DN100
Function	Space heating & DHW supply	Mixed thermal return, Source for central heat pump	Space cooling supply

3.5 Network topology and layout

3.5.1 Distribution network architecture

The proposed three-pipe network follows a branched tree topology, which is the most common architecture for district heating energy distribution due to its simplicity and ease of sectional isolation. The central energy plant is located at the root of the distribution tree, which the three pipes radiate outward to serve individual building substations at the branch terminals.
The pipe alignment adopted for this thesis has a single trench, three-pipe bundle. Pipe 1(hot supply at 60°C) and Pipe 2 (common return at 18-23°C) are installed together within a shared twin casing while Pipe 3 (cold supply at 6°C) is installed within its own separate casing, apart by a defined trench gap. Both casings are buried at a depth of 0.8-1.2 meters below ground level in accordance with the standard district heating installation practice in Nordic countries (Frederiksen & Werner, 2013). This split-casing arrangement retains most of the excavation and footprint savings of co-routing since hot and return pipes, which have the smallest temperature differential share a single casing which reduce risk of inter-pipe heat transfer between Pipe 1 and Pipe 3 by physically separating them into independent casings.
For the purposes of the analytical model in this thesis, the network is considered as a single pipe segment of 5000 m in length, connecting the central plant to a terminal substation. This length is chosen as representative of a typical urban distribution segment and is consistent with the modelling approach adopted by (Xu, et al., 2021)in their heat loss analysis. Pressure drop calculations and pump sizing are not primary focus of this study, but representative values for pump head and the flow velocity to establish the completeness of the system definition.

3.5.2 Pipe alignment and casing configuration

Figure 2: Pipe layout. shows the trench cross-section adopted for the proposed system. Pipe 1 (H) and Pipe 2 (R) are carried within a shared twin casing of outer diameter (Da) 225 mm, with a centre-centre spacing between the H and R pipe cores of c12=68 mm. Pipe 3 (C) is carried within a separate, independent casing, also of outer diameter (Da)=225 mm. The two casings are set apart such that the edge-to-edge trench gap between the shared twin casing and the separate cold casing is 150 mm, giving a centre-to-centre distance between the twin casing and the cold casing is 150 mm. This is primary structural measure used to control inter-pipe heat

transfer between the hot supply and the cold supply. Moreover, pipe 2 carries the combined heating and cooling return flow as shown by the mass flow rates derived in Chapter 4. This can substantially exceed the flow, the DN65 pipe shown in Table 1: Design parameters for each pie in the proposed three-pipe system. has been considered for consistency with the casing geometry, but a full hydraulic design would be needed to verify this against pressure drop limits and combined flow velocities and may require pipe 2 to be sized independently rather than matched with pipe 1. In a real installation, the common return pipe would not be sized as a constant diameter either. Moving in the direction of flow toward the central plant, pipe 2 collects return flow from heating and cooling substations, so the mass flow increases progressively along the route, and therefore the required diameter increases progressively along the route, with a step increase occurring specifically at the point where the cooling return joins the heating return main. This is the reverse of the tapering process that would apply on the supply side.

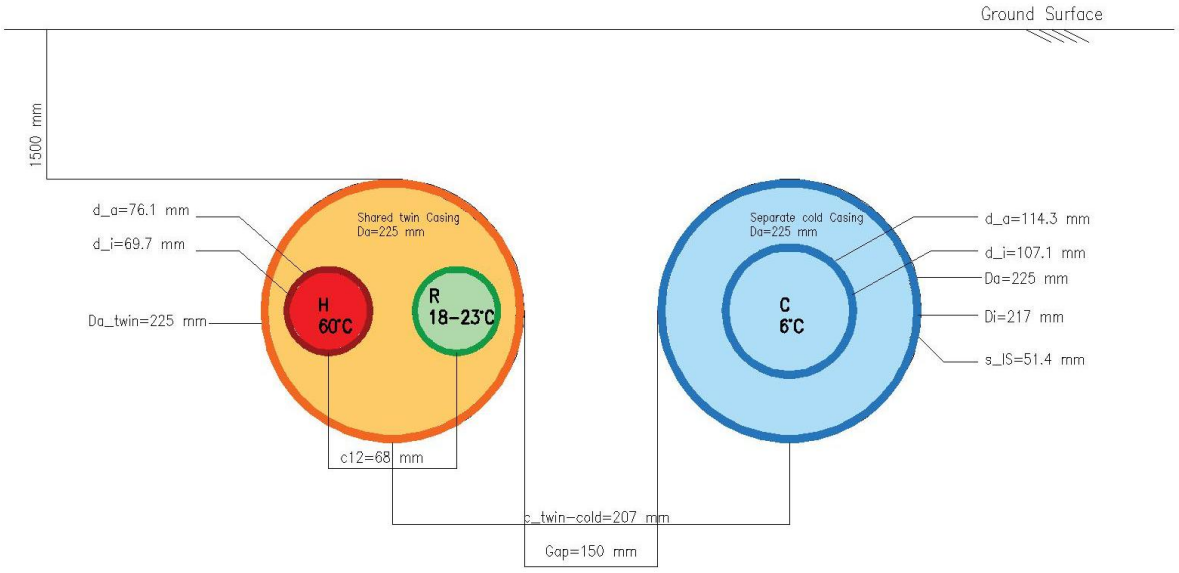


Figure 2: Pipe layout.

Table 2:Pipe alignment and casing dimensions.

Parameter	Value (mm)	Notes
Shared twin casing outer diameter (Da)	225	Contains Pipe 1 and Pipe 2
Twin casing centre-to-centre spacing (c12)	68	Between Pipe 1 and Pipe 2 within twin casing
Separate cold casing outer diameter (Da)	225	Contained Pipe 3
Trench gaps between casing	150	Edge to edge, twin casing to cold casing
Centre-to-centre distance, twin casing to cold casing (twin-cold)	207	Govern inter-pipe heat transfer between Pipe 1 and Pipe 3

3.5.3 Substation connection principle

Building substation connect to the three-pipe network through a parallel connection arrangement, in which each substation taps into Pipe 1 (if it has a heating demand), Pipe 3 (if it has a cooling demand), or both. The return flows from each substation are combined and discharged into Pipe 2 through individual return connection points.

The substation heat exchanger in the heating side is a plate heat exchanger that transfers heat from the Pipe 1 supply flow to the secondary building heating circuit, with the primary side return entering Pipe 3. The substation heat exchanger on the cooling side similarly transfers cooling capacity from the Pipe 2 supply to the secondary building cooling circuit, with the primary return also entering Pipe 3. This arrangement ensures that Pipe 2 receives both return streams at all substations simultaneously, enabling the mixing dynamic that is central to the system's thermodynamic advantage.

A critical design requirement for the substation is the hydraulic separation of the heating and cooling return flows prior to their discharge into Pipe 3. Without separation, the warm heating return flow and the cool cooling return flow could mix within the substation itself, creating temperature peaks that reduce the ΔT on both circuits. The preferred approach is to connect the two return streams to Pipe 2 at the same tee junction but through separate non-return valves allowing mixing to occur within the main return pipe rather than within the substation.

3.6 Central energy plant configuration

3.6.1 Plant layout and equipment

The central energy plant is the production and distribution hub of the three-pipe system. In the proposed configuration, it comprises a central heat pump unit responsible for both heating and cooling system together with a Y-separator fitting that divides the incoming Pipe 2 mixed return flow between the evaporator and condenser of the heat pump. The heat pump operates on the vapour compression cycle and use mixed return flow from Pipe 2 as the thermal interface with the distribution network.

The Y-separator divides the Pipe 2 mixed return stream, which arrives at the plant at the calculated mixed return temperature of approximately 18-23°C, into two parallel sub-streams of proportioned mass flow. The first stream is routed to the evaporator side of the central heat pump, where it is cooled from the mixed return temperature down to the design cold supply temperature (T cold) of 6°C, this stream is then discharged into Pipe 3 and distributed to the district cooling network. The second sub-stream is routed to the condenser side of the central heat pump, where it receives the heat rejected but the refrigerant circuit and is heated up to the design hot supply temperature (T hot) of 60°C, is then discharged into Pipe 1 and distributed to the district heating network.

Since a single machine simultaneously produces both the district heating and district cooling supply streams from a common source, the heat pump requires a temperature lift from approximately 18-23°C to 60°C, where the evaporator side requires a temperature depression from approximately 18-23°C to 6°C. Both figures are smaller than the lifts implied by using ambient air, which is the principal thermodynamic benefit of coupling the heat pump directly to the shared return pipe. The coefficient of performance (COP) of the heat pump depends on the relative magnitudes of the heating and cooling loads and is discussed in Chapter 4.

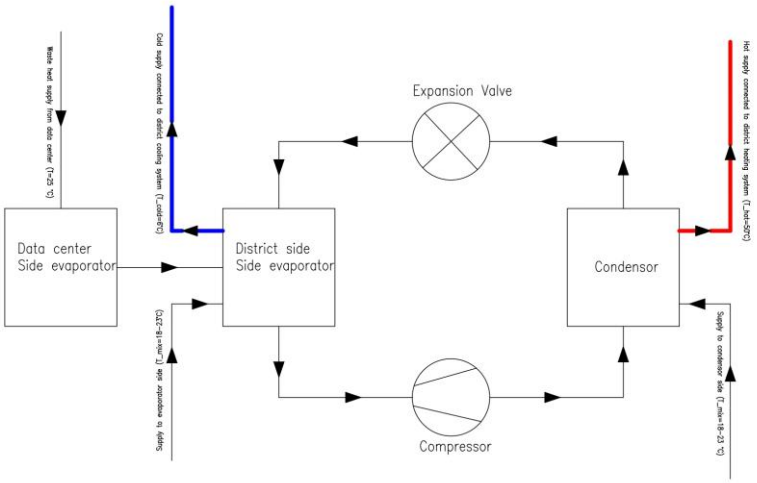


Figure 3: Heat pump diagram.

3.7 Dual-evaporator configuration and data centre waste heat recovery

A distinguishing feature of the plant design is that central heat pump is fitted with two separate evaporator circuits rather than one, allowing it to draw simultaneously on two independent heat sources of different character. The first evaporator circuit referred to as the district side evaporator supplied with the Y-separator sub-stream of the Pipe 2 mixed return, at approximately 18-23°C. This circuit performs a dual function; it supplies heat to the evaporator side and condenser side of the central heat pump. The second circuit referred to as the data centre side evaporator is supplied with waste heat recovered from the data centre cooling loop, arriving at the central plant at approximately 25°C. Unlike the district-side evaporator, this circuit is not a part of the three-pipe distribution network and does not produce a supply stream for building substations, the sole function of this circuit is to act as an additional heat source.

On the refrigerant side, the two evaporator circuits are connected to a common compressor suction line, so that the vapour generated in each evaporator is combined within the machine before compression rather than two water streams mixing directly. This allows the heat pump controller to adapt two sources according to their availability and temperature. The district-side evaporator approximately between 18-23°C and data centre-side typically between 25°C, presents a smaller lift temperature lift than district-side and therefore can be prioritised where its heat flow is sufficient to meet the demand, with district-side evaporator making up the balance of the demand. The relative contribution of the two evaporator circuits to the total demand, and its effect on overall coefficient of performance (COP) is analysed in chapter 4.

The addition of the data centre waste heat source improves the overall thermodynamic performance of the plant in two ways. First, because the data centre stream is available at a higher and comparatively stable temperature than the mixed district return, it reduces the average compressor lift required to reach the 60°C hot supply, particularly under conditions when the district mixed return temperature is at the lower end of its 18-23°C range. Second, because the data centre heat source is largely dependent of the district heating and cooling load patter, it provides a heat input that persist through periods when district-side evaporator load is reduced by low simultaneous heating and cooling demand, improving the availability and reliability of the central plant output.

3.8 Y-Separator operation and continuous heat recovery

A key operational characteristic of the plant configuration is that heat recovery between the cooling and heating is continuous and rather than normal operation, it can be controlled. Because both evaporator sub-stream and the condenser sub-stream originate from the same Pipe 2 mixed return and are processed by the same central heat pump. Every unit of heat extracted from the evaporator sub-stream to produce 6°C cold supply, together with the compressor work input, is delivered to the condenser sub-stream to produce 60°C hot supply. The electrical energy consumed by the heat pump compressor is the only primary energy input to this process. And the effective system efficiency expressed as the ratio of total useful thermal output to electrical energy consumed. This efficiency can exceed the achievable with a conventional system that uses separate heat production.

The Y-separator split ratio between the evaporator sub-stream and the condenser sub-stream is governed by the relative instantaneous heating and cooling demand across the customers and controlled by modulating valves and separate outlets. Under peak heating conditions, a larger proportion of the mixed stream is routed to the condenser side, likewise, under peak cooling conditions, a larger proportion is routed to the evaporator side. During the shoulder season simultaneous load scenario, the split approaches a balance between the demands.

When cooling is delivered to a customer, heat is removed from the building and added to the network. When heating demand is greater than the heat added by the cooling customers, the system has a heat deficit. This deficit is created by heat delivered to heating customers (together with pipe heat losses). It is covered by the waste heat from the data centre and by the electricity used by the compressor, which is converted to heat in the heat pump. The energy balance of the heat pump is:

$$\text{Heat delivered by the condenser} = \text{heat taken from the district side evaporator} + \text{heat taken from the data centre} + \text{compressor electricity}$$

Since the heat taken from the district side evaporator is the cooling delivered, the heating delivered minus the cooling minus the cooling delivered equals the data centre heat plus the compressor electricity (losses are neglected).

3.9 Inter-Pipe heat transfer and mitigation

3.9.1 Nature of the problem

The large temperature difference between Pipe 1 (heating supply at 60°C) and Pipe 3 (cooling supply at 6°C), approximately 54°C, creates a conductive heat transfer through the soil wherever these two pipes are installed in proximity. This inter-pipe heat transfer is not controlled may leading to the progressive warming of the cooling supply along the pipe route. This means that the cooling supply temperature at the downstream end of the network may be measurably higher than at the central plant. In a poorly designed system, this warming can reduce the cooling capacity available at the end customers, increase the lift required at the evaporator side of the central heat and raise energy consumption. (Arabkoohsara, et al., 2019) quantified this effect using CFD modelling for a bundled three-pipe geometry and found that inter-pipe heat transfer rates are highly sensitive to the separation distance between the pipe centrelines and the thermal conductivity of the surrounding soil. Their results indicate that a centroid separation of at least 300-400 mm between the heating and cooling pipes, also with enhanced insulation on the cooling supply pipe is necessary. Because Pipe and Pipe 3 have the largest temperature differential of any pipe pair in the system., the pipe alignment adopted specifically configure to keep them apart, rather than relying on insulation alone.

3.9.2 Mitigation strategies

The primary mitigation strategy adopted in the proposed system design is separation through the split casing pipe alignment. Pipe 1 and Pipe 2 which have a small temperature differential of approximately 30°C are accommodated together within a single shared twin casing, at a close centre-to-centre spacing of (c12) equal to 68 mm, since the inter-pipe heat transfer between them is limited and heat lost from the hot pipe is largely absorbed by the return pipe rather than lost to surrounding soil. Pipe 3 which has the largest temperature differential relative to Pipe 1 (approximately 54°C) is instead installed in its own separate casing, set apart from the twin casing by a 150 mm trench gap, giving a centre-to-centre distance of twin-cold=207 mm between the twin casing and cold casing. This separation is comfortably within the range identified by (Arabkoohsara, et al., 2019) as necessary to limit cooling supply temperature degradation and removes Pipe 3 from the direct soil conductive path of the hot supply entirely. As second mitigation strategy, applied in addition to the casing separation, is to enhance insulation on Pipe 3. While Pipe 1 and Pipe 2 are specified with standard pre-insulated pipe insulation (Series 2 and Series 1 respectively) within the shared twin casing. Pipe 3 is specified with a Series 3 enhanced insulation jacket within its independent casing, which provides approximately 50% greater insulation thickness than the standard specification. Also, within the shared twin casing itself, no additional buffering is required between Pipe 1 and Pipe 2, since heat transfer between the hot supply and the common return does not represent an uncontrolled loss to the environment.

3.10 Comparison with the conventional Four-Pipe system

Table 3:Structural comparison between the proposed three-pipe system and conventional four-pipe system. presents a direct structural comparison between the proposed three-pipe system and the conventional four-pipe district heating and cooling network. The comparison is restricted to system definition and design parameters, efficiency and cost comparison.

Table 3:Structural comparison between the proposed three-pipe system and conventional four-pipe system.

Design Attribute	Three-pipe system	Four-pipe system
Number of distribution Pipes	3	4
Heating supply pipe	Pipe 1 at 60 °C	Heating supply at 60-80°C
Cooling supply pipe	Pipe 3 at 6 °C	Cooling supply at 6-10°C
Heating return pipe	Shared return (Pipe 2)	Dedicated return pipe
Cooling return pipe	Shared return (Pipe 2)	Dedicated return pipe
Return temperature	Mixed: load dependent (18-23°C)	Independent
Central plant heat source/sink	Mixed return feed	Separate source (heating) and sink (cooling)
Inter-pipe heat transfer	Moderate	Low
Heat recovery potential	High, continuous	Low
Trench width	Reduced width	Standard
Substation Complexity	Moderate	Standard
Central plant configuration	Dual-evaporator central heat pump with Y-seperator on shared return, and second evaporator on data centre waste heat (25°C)	Separate boiler/chiller unit
Secondary heat source	Data centre waste heat loop (25°C) independent of district load	None

3.11 Design assumptions and limitations

3.11.1 Steady-state assumption

The analytical model in this thesis assumes steady state operating within each defined scenario. This means that the temperatures, flow rates and heat transfer rates at each point in the network are assumed to be constant over the duration of each scenario. This assumption is standard in first order district energy system analysis and is consistent with the approach adopted by (Xu, et al., 2021) and (Arabkoohsara, et al., 2019) in their three-pipe studies.
The steady state assumption introduces a known limitation: in real systems, the load ratio between heating and cooling consumers varies continuously, causing the mixed return temperature in Pipe 2, and consequently the Y-separator split ratio, to fluctuate dynamically. The three defined scenarios (peak heating, peak cooling, and simultaneous load) are used to analyse the range of likely operating conditions and provide conservative and optimistic performance estimates for each relevant metric.

3.11.2 Assumed load profiles

In the absence of measured building energy demand data for a specific site, the load profiles used in this are based on a per house load of 23 kW, calculated using our previous work on district heating assignments. For this thesis, 10 houses are considered, resulting in a total district heating and cooling demand of 230 kW each. No outdoor temperature assumptions were made for any of the load scenarios, the heating and cooling values derived from the per-house load calculations.

3.11.3 Pipe and soil thermal properties

The thermal properties of the pipe insulation and the surrounding soil used in the heat loss model are drawn from the values reported in (Xu, et al., 2021) (Arabkoohsara, et al., 2019) Logstor pipes (LOGSTOR, 2025), Table 4: Assumed thermal properties of pipe and soil materials. summarises the key assumed thermal property values.

Table 4: Assumed thermal properties of pipe and soil materials.

Parameter	Value	Source
Soil thermal conductivity	1.5 W/m.K	(Xu, et al., 2021)
Annual mean soil temperature	8°C	Swedish mean value
Pipe burial depth	1.0 m	logstor
Pipe insulation conductivity (PUR foam)	0.027 W/m.K	logstor
Pipe Inner diameter- Pipe 1	DN65	(Arabkoohsara, et al., 2019)
Pipe Inner diameter- Pipe 2	DN65	(Arabkoohsara, et al., 2019)
Pipe Inner Diameter-Pipe 3	DN100	(Arabkoohsara, et al., 2019)
Twin casing outer diameter (Pipe 1 and Pipe 2)	225 mm	Pipe alignment drawing
Cold casing outer diameter (Pipe 3)	225 mm	Pipe alignment drawing
Centre-to-centre spacing, Pipe1-Pipe2 (c12)	68 mm	Pipe alignment drawing
Centre-to-centrecenter, twin casing-cold casing	207 mm	Pipe alignment drawing
Trench gap, twin casing-cold casing	150 mm	Pipe alignment drawing

3.12 Strengths and limitations of the proposed system

3.12.1 Strengths

The proposed three-pipe system offers several structural and thermodynamic strengths compared to the conventional four-pipe alternative. First, the elimination of one pipe reduces capital expenditure across all cost categories, pipe material, insulation, trench volume, and substation connections. This makes three-pipe configuration more economically attractive for a new build energy project. Secondly, the shared return pipe combined with a Y-separator and a single central heat pump, enables both the heating and cooling from moderate temperature mixed return flow rather than an ambient air or ground. This moderate source/sink temperature reduces the temperature lift required for heating and temperature depression for cooling, thus improving the coefficient of performance (COP) of the machine and reducing electrical energy consumption.

Thirdly, the heat recovery potential of the shared return through a single central heat pump represents a continuous energy saving that is not available in a four-pipe system without dedicated heat recovery system. In the proposed configuration, this heat recovery occurs without the need for additional equipment beyond the central heat pump and Y-separator that would be required in any 4GDH plant. Finally, the dual-evaporator configuration allows the central heat pump to draw on a second external low grade waste heat from data centre as a heat source approximately at 25°C. Because this second source is largely isolated from the seasonal pattern of the district heating and cooling load, it improves the availability and average of coefficient of performance (COP) of the central heat pump and thus improving primary energy saving opportunity.

3.12.2 Limitations and challenges

The proposed system also presents several design and operational challenges that must be acknowledged. The most significant is the loss of independent control over the heating and cooling return temperatures. In a conventional four-pipe system, the return temperatures of the heating and cooling circuits can be managed independently by adjusting substation control settings. In the three-pipe system, the mixed return temperature in Pipe 2 is an emergent property of the simultaneous load conditions, and it cannot be independently set without restricting the flows from either the heating or cooling circuit, or without adjusting the Y-separator split ratio. This loss of control independence complicates the operation of the central heat pump plant under conditions where the mixed return temperature deviates significantly from the design value.

Hydraulic complexity is a further challenge. The convergence of two return streams at each substation connection point, and their division at the Y-separator, requires careful pressure balancing to ensure that flows from heating substation and cooling substation enter Pipe 2 without creating pressure differentials that could reverse flow in one of the circuits. And y-separator can proportion flow to the evaporator and the condenser sides without starving either heat pump component. This requires more sophisticated hydraulic modelling and control valve specification than a conventional two-pipe or four-pipe system. Control system complexity is higher, as the central plant must continuously monitor and respond to the variable mixed return temperature and the balance between heating and cooling loads.

Finally, the seasonal demand variation in any district with both heating and cooling loads means that the thermodynamic advantage of the shared return pipe, and the continuous heat recovery benefit, are also variable. However, this benefit is higher during shoulder seasons and

decreases during peak winter and peak summer conditions when the load is dominated by a single circuit. The system must therefore be designed to perform acceptably across the range of seasonal conditions, not only under the shoulder season but also peak load scenarios.

4 Methodology

This chapter presents the methodological framework used to evaluate the performance of the proposed three-pipe fourth-generation district heating and cooling (4GDHC) system. The methodology is grounded in analytical thermodynamic modelling, drawing on energy balance principles, heat transfer theory, and heat pump performance relations to quantify the key performance indicators (KPIs) identified in the research questions in Chapter 1.

The research strategy adopted in this thesis is a theoretical and model-based investigation because the proposed system configuration does not exist as a physical installation and therefore cannot be evaluated through a direct experimental method. This approach is consistent with the methodologies in analogous studies of novel district energy topologies, including the three-pipe heat loss modelling of (Xu, et al., 2021),and the CFD analysis of (Arabkoohsara, et al., 2019).

4.1 Research approach and justification

Analytical modelling was selected because of three reasons. First, the research questions are fundamentally quantitative in nature: they require numerical estimates of temperature differentials, coefficient of performance (COP) values and heat transferred rates to be answered meaningfully.

Second, the system under study is a novel configuration for which no data exist. Analytical modelling provides a theoretically rigorous means of characterising system behaviour under input conditions, and its results can serve as the basis for the design of future experimental or simulation-based validation studies.

Third, analytical models offer transparency and reproducibility that are essential qualities in an academic research context. All equations, assumptions, and parameter values stated in this chapter enable independent verification of the results and straightforward substitution of alternative input data for site-specific applications.

4.2 Pipe heat transfer model

This section represents the complete thermal resistance used to compute the heat loss of each buried pipe and the heat transfer between them. Each pipe loses or gains heat through three resistances in series, from the fluid to the inner pipe wall, through the pipe wall itself, and from the outer surface to the soil. The model follows the buried multi-pipe method described by (Frederiksen & Werner, 2013)and the mutual interference superposition approach used by (Xu, et al., 2021)for the three-pipe district heating network.

4.2.1 Pipe wall thermal resistance

Each pipe is a standard pre-insulated district heating pipe, made up of a steel conduit, a layer of rigid polyurethane foam insulation, and an outer casing. The thermal resistance of the pipe wall, R_{pipe} is the sum of the resistances of each of these three layers.

$$R_{pipe} = RST + RIS + RSH \qquad Eq.(4.1)$$

$$RST = \left(\frac{1}{2\pi kST}\right) \times \ln\left(\frac{d_a}{d_i}\right) \qquad Eq.(4.2)$$

$$RIS = \left(\frac{1}{2\pi kIS}\right) * Ln\left(\frac{d_i}{d_a}\right) \qquad Eq.(4.3)$$

$$RSH = \left(\frac{1}{2\pi kSH}\right) * Ln\left(\frac{d_a}{d_i}\right) \qquad Eq.(4.4)$$

Where:

RST - heat transfer resistance through the conduit steel pipe.
RIS – heat transfer resistance through insulation.
RSH – heat transfer resistance through the shield pipe.
KST, KIS, KSH– thermal conductivity of steel service pipe, polyurethane insulation, and high-density polyethene outer casing, respectively [W/m. K]
Di, Da – inner and outer diameters of the outer casing [m]

4.2.2 Soil thermal resistance

Heat loss from each buried pipe to the surrounding soil is modelled using the analytical solution for a buried cylinder with uniform surface temperature, based on the method (Frederiksen & Werner, 2013) and adopted (Xu, et al., 2021) for three pipe networks. Two forms of the equation apply, depending on the ratio of burial depth to outer pipe diameter:

$$R_{gr} = \frac{1}{2\pi K_{gr}} Ln\left[\frac{2H_m}{D_a} + \sqrt{4\left(\frac{H_m}{D_a}\right)^2 - 1}\right], for \frac{H_m}{D_a} < 2.0 \qquad Eq.(4.5)$$

$$Rgr = \left(\frac{1}{2\pi kgr}\right) * Ln\left(\frac{4Hm}{Da}\right), for \frac{Hm}{Da} \geq 2.0 \qquad Eq.(4.6)$$

Where:
k_{gr} - thermal conductivity of surrounding soil [W/m.K].
D_a – Outer diameter of the pipe
H_m— effective burial depth.

$$H_m = H + H_f \qquad Eq.(4.7)$$

$$H_f = \frac{K_{gr}}{h_a} \qquad Eq.(4.8)$$

Where:
H - physical burial depth H
H_f - equivalent air-film depth correction
h_a - air-to-ground surface heat transfer coefficient.

4.2.3 Mutual thermal interference between pipes.

Since the three pipes are buried close together, heat conducted from one pipe toward the surface is partially taken by the adjacent pipes, creating a thermal interaction. For each pair of pipes a mutual interference resistance is defined:

$$R_{(ij)} = \frac{1}{2\pi K_{gr}} Ln\left[1 + \sqrt{4\left(\frac{H_m}{C}\right)^2}\right] \qquad Eq.(4.9)$$

Where C_{ij} is the centre-to-centre spacing between pipe i and pipe j. Three such terms are required for a three-pipe system

$R_{(12)}$ — hot supply - common return

$R_{(13)}$ — hot supply - cold supply

$R_{(23)}$ — common return - cold supply

4.2.4 Convective resistance at the pipe wall

The resistance to heat transfer between the flowing water and the inner pipe wall, R_h depends on the flow regime and is calculated from the convective heat transfer coefficient h_f:

$$R_h = \frac{1}{\pi * h_f * d_i} \qquad Eq.(4.10)$$

$$h_f = \frac{Nu * K_f}{d_i} \qquad Eq.(4.11)$$

WhereK_f – is the thermal conductivity of water, and Nu being the Nusselt number, obtained from the Dittus-Boelter correlation for turbulent flow or taken as a constant for laminar flow.

4.2.5 Heat loss by Three-pipe superposition

The net heat loss per unit pipe length for each of the three pipes is obtained by superposing the pipes own conduction path to the ground with two mutual interference correction terms accounting for heat exchanged with its neighbours:

$$q_1 = \frac{T_1 - T_E}{\sum \frac{R_1 - (T_1 - T_2)}{R_{12} - (T_1 - T_3)/R_{13}}} \qquad Eq.(4.12)$$

$$q_2 = \frac{T_2 - T_E}{\sum \frac{R_2 - (T_2 - T_1)}{R_{12} - (T_2 - T_3)/R_{23}}} \qquad Eq.(4.13)$$

$$q_3 = \frac{T_3 - T_E}{\sum \frac{R_3 - (T_3 - T_1)}{R_{13} - (T_3 - T_2)/R_{23}}} \qquad Eq.(4.14)$$

where:

T_1 – Hot supply temperature

T_2- Common return temperature

T_3- Cold supply temperature T_E

T_E - undisturbed average ground temperature

$\sum R_i = R_{h,i} + R_{pipe,i} + R_{gr,i}$ is the total resistance from the fluid in pipe i to the ground surface

4.2.6 Direct Hot-to-Cold heat transfer and cooling degradation.

The direct heat transfer between the hot supply and cold supply through the intervening soil is defined as:

$$q_{1-3} = \frac{T_1 - T_3}{R_{13}} \qquad Eq.(4.15)$$

This term represents the fraction of the total cold pipe heat gain that comes specifically from the hot pipe as opposed to the common return pipe or the undistributed soil, and is used together with the pipe route length L and the cooling circuit mass flow rate to estimate the temperature rise of the cooling supply fluid along the pipe route:

$$T_{gain} = \frac{q_3 * L}{\dot{m} * c_P} \qquad Eq.(4.16)$$

The temperature rise represents the degradation of the cold supply temperature at the downstream end of the network relative to the plant outlet temperature.

4.3 Temperature drop in the direction of flow

While the section above evaluates the heat loss per unit pipe length at a given temperature, that heat loss causes the fluid temperature to drop itself slowly as it flows along the pipe. This method tracks the temperature drop for both hot supply and cold supply for the range of pipe diameters DN65 – DN150.

4.3.1 Energy balance derivation

Consider a differential element of pipe of length d_x. The steady-state energy balance on the fluid states that the heat lost through the pipe wall to the surroundings equals the enthalpy change of the following fluid.

$$\dot{m}c_p\Delta T = -U.(\pi d_o).(T - T_a)d_x \qquad Eq.(4.17)$$

Where:

$\dot{m} = \rho v A_i = \rho v \frac{\pi d_i^2}{4}$ - mass flow rate [kg/s]

U = K – overall heat transmission coefficient per unit outer pipe area [W/m^2k], combining pipe wall, insulation and soil resistance.

d_o – outer pipe diameter [m]

T – fluid temperature at position x [°C]

T_a -Ambient ground temperature [°C]

Substituting $\dot{m}$ and rearranging gives the formula:

$$\frac{dT}{dx} = -\frac{4Kd_o(T - T_a)}{vd_i^2\rho c_p} \qquad Eq.(4.18)$$

For practical calculations, the differential relationship is applied over short pipe segments of length L 100m in this study. Assuming T is approximately constant (equal to segments inlet temperature). A standard finite-difference (Euler) approximation valid provided L is short relative to the overall decay length scale of the exponential temperature profile:

$$\Delta T_S = -\frac{4KLd_o(T_s - T_a)}{vd_i^2 \rho c_p} \qquad Eq.(4.19)$$

The calculation is done segment by segment over a total length of 5km, where hot supply starts at 60°C and cold supply starts at 6°C.

4.4 Mixed (Common) return temperature model

The temperature of the fluid in the common return at the central plant inlet is determined by the mixing of the mass flow rate of heating return flow and the cooling return flow. Assuming steady-state conditions, the mixed return temperature T_{mix} is given by the following energy balance:

$$T_{mix} = \frac{(m_h * T_h) + (m_c * T_c)}{m_h + m_c} \qquad Eq.(4.20)$$

Where:
m_h - mass flow rate of the heating circuit return
m_c - mass flow rate of the cooling circuit return
T_h - Temperature of the heating circuit return
T_c - Temperature of the heating circuit return

The mass flow rate m_h and m are derived from the thermal loads and the design temperature differentials of each circuit.

$$\dot{m} = \frac{\dot{Q}}{c_p * (T_s - T_r)} \qquad Eq.(4.21)$$

Where:
$\dot{Q}$ – Heat/cooling load [kW]
c_p – Specific heat capacity of water [J//kg K)]
$T_s \; and \; T_r$ - supply and return temperatures [°C]

4.4.1 Temperature differential calculation

The effective temperature difference (ΔT) of the distribution system is the key performance indicator for the energy network. It governs the mass flow rate required to deliver the given thermal load and thereby determines pumping energy consumption and pipe size requirements. Moreover, it is the most directly relevant to heat pump performance and energy efficiency.

4.5 Heat pump performance model

Two independent methods are used to estimate the coefficient of performance coefficient of performance (COP) of the central heating and cooling plant (simultaneous heating and cooling heat pumps): the Carnot method and Lorenz mean temperature method, which accounts for the temperature glide of the refrigerant across the evaporator and condenser. Both methods are applied to the same set of loads and mixed return temperatures so that the two estimates can be compared directly.

4.5.1 Carnot method

The heat exchanger effectiveness formula (ε) is used to find the refrigerant's evaporating and condensing reference temperatures away from the actual water temperatures. (Bergman, et al.,

n.d.)

$$\varepsilon = \frac{T_{water,in} - T_{water,out}}{T_{water,in} - T_{refrigerant}} \qquad Eq.(4.22)$$

A single reversible heat-transfer assumption within each heat exchanger (isothermal evaporation and condensation), consistent with $C_{min} = C_{water}$ for the water side, since the refrigerant changes phase at effectively constant temperature while it is evaporating or condensing.

$$COP_{heating} = \frac{T_{ref,condenser}}{T_{ref,condenser} - T_{ref,evoprator}} \qquad Eq.(4.23)$$

$$COP_{cooling} = \frac{T_{ref,evoprator}}{T_{ref,condenser} - T_{ref,evoprator}} \qquad Eq.(4.24)$$

4.5.2 Lorenz mean-temperature method

Since the Carnot method still assumes the refrigerant temperature is constant across the heat exchanger, it does not capture the fact that a real refrigerant is superheated on leaving the evaporator and subcooled on leaving the condenser, both of which change the temperature at which heat is exchanged. The Lorenz method addresses this by using the logarithmic mean temperature of the refrigerant across each heat exchanger and by applying an explicit efficiency η of 0.45 to 0.6 (Lund, et al., 2017)to the resulting ratio.

$$COP = \eta * \frac{T_{High}}{T_{High} - T_{Low}} \qquad Eq.(4.25)$$

$$T_{high}\ or\ T_{low} = \frac{T_{in} - T_{out}}{Ln(T_{in}) - Ln(T_{out})} \qquad Eq.(4.26)$$

Here, T_{in} and T_{out} are the inlet and outlet temperatures of the condenser and the evaporator in the heat pump.

4.5.3 Source and sink temperature

For the central heating heat pump. the source temperature is the mixed return temperature T_{mix} at the central plant inlet, and the sink temperature is the hot supplied temperature. For the central cooling heat pump, the source temperature is the mixed return temperature T_{mix} and the sink temperature is the cold supplied temperature. This arrangement means that both heating and cooling coefficient of performance (COP) are functions of T_{mix} and both improve when T_{mix} sits in the moderate range achievable under the simultaneous load scenario.

4.5.4 System energy balance

The energy balance of the system is considered at the system boundary, where it can be seen as heat gains and heat losses only. The gains are the electricity supplied to the heat pump compressor the heat gains such as, the cooling delivered to the customers, because heat taken from their buildings is returned to the network, and the auxiliary heat supplied to the system, here from data centre. Moreover, any heat transferred to the cold supply pipe or from the supply pipes to the return in the distribution network is also a gain. The losses are the heat delivered to the customers and the heat lost from the hot supply pipe to the ground. In steady state the total heat gains equal to the total heat losses.

$$Electricity + Cooling\ delivered + Auxiliary\ heat + Heat\ gain\ in\ the\ distribution = Heat\ delivered + Heat\ loss\ in\ distribution \quad Eq.(4.27)$$

A heat pump does not create heat, it moves heat from a source to sink, so external heat input is required. In this system, heat is taken from an auxiliary source and from the cooling return. Heating and cooling demand should therefore be of similar size to keep the balance easy to maintain, and a chiller or other heat rejection may be needed when the balance cannot be met. The 230KW case is an isolated reference case used only to determine the mixed return temperature and the COP.

4.6 Reference systems and economic comparison

To evaluate the economic feasibility of the proposed three-pipe simultaneous heating and cooling distribution concept against two benchmark alternatives, which are the conventional four-pipe heating and cooling distribution system and the three-pipe system at Ranagård, Sweden. A cost-per-meter analysis is carried out for these three distribution layouts.
Following standard district heating network costing practice, the total installed cost per meter of pipeline is broken down into five cost categories:

$$C_{total} = C_{material} + C_{civil} + C_{design} + C_{permits} + C_{labour} \quad Eq.(4.28)$$

The categorical adjustment factor has been taken from (Energiföretagen, 2022), which defines four area-based cost categories reflecting site installation conditions:

Table 5:Four area-based cost categories reflecting site installation conditions.

Category	Swedish term	Description	Factor
A	Innerstad	Inner-city areas with dense, pre-existing buried infrastructure and hard surfacing (asphalt, paving); typically, the highest-cost category due to reinstatement work, utility conflicts, and traffic management	1.3
B	Ytterområde	Peripheral/suburban areas with more variable infrastructure density and surfacing	0.9
C	Parkmark	Park and green-space areas; lower and more uncertain cost basis	0.7
D	Exploitering	Pipe laid in conjunction with new-build development	0.5

In this study, Category A (factor 1.3) is applied, reflecting an assumed inner-city/dense urban installation route for the case-study network

4.6.1 Application to the three configurations

The same five- category cost structure is applied to all three configurations with the pipe material, pipe work, casing category being the primary driver of difference between them since it directly reflects the number and size of the pipe required.

Table 6:Three configurations and pipe basis assumed.

Configuration	Pipe basis assumed
Three-pipe system	Hot supply (DN65) + common return (DN65) + cold supply (DN100)
Four-pipe system	Independent heating twin-pipe (DN65 supply + return) and independent cooling twin-pipe (DN100 supply + return)
Ranagård system	Twin-pipe system per [source], reported as DN125 outer-diameter-equivalent twin pipe, with a DN40 recirculation pipe, used here as a single-service (heating-only) cost benchmark

The base unit cost rates (per meter of pipeline, by nominal diameter DN20–DN700, broken down into pipe/civil/design/permitting/labour sub-categories) are taken from: (Energiföretagen, 2022).

5 Results and Analysis

5.1 Heat loss during distribution

The first analysis looks at what happens to the thermal performance, using the resistance network superposition method with the hot supply at 60°C, the common return at 20°C and the cold supply at 6°C. In this the hot supply and the common return are held together in a shared casing, while the cold supply is laid separately in its own casing nearby.

Table 7:Results of thermal performance.

Quantity	Value	Description
$R_{pipe1} = R_{pipe2}$	3.355 mK/W	DN 65 twin pipe (Hot supply + Return pipe)
R_{pipe3}	3.796 mK/W	DN100 (cold supply)
$R_{gr1} = R_{gr2}$	0.374 mK/W	Soil resistance (Shared twin casing)
R_{gr3}	0.311 mK/W	Soil resistance cold casing (separate, large casing)
$R_{twin} - cold$	0.263 mK/W	Mutual interference resistance between the twin-casing centre and cold pipe

The twin pipe number is good for a well-insulated pipe pair running at 60°C and 20°C, buried in ground which is 8°C had a loss of 8.02 W/m, which is a reasonable value for a modern district heating system. For the cold pipe, which carries 6°C water through ground at approximately 8°C, the temperature difference is only 2 K, providing very little driving force for heat exchange with the soil. Therefore, the remaining heat gain, effectively 128.89 W/m, comes from the hot supply line. This calculated value should be interpreted as a thermal penalty associated specifically with the shared trench configuration.

The magnitude of this interference term is governed by the heating side operating temperature and the physical separation between the two casings. Therefore, increasing the spacing between cold pipe and twin casing would reduce the heat loss. It should be presented as a design parameter available for future optimisation.

5.2 Temperature drop along distribution route

Separately from the heat loss during distribution, which has been discussed above, the pipe temperature drop was calculated for pipe sizes from DN65 to DN150 over 5km length. Using an overall heat transmission coefficient, which was varied by pipe size (ranging from 0.38 down to 0.22 W/m²K on the heating side and 0.32 down to 0.18W/m²K on the cooling side).
shows the resulting temperature profile along the heating supply pipe, and pipe shows the corresponding profile along the cooling supply pipe, for each diameter.

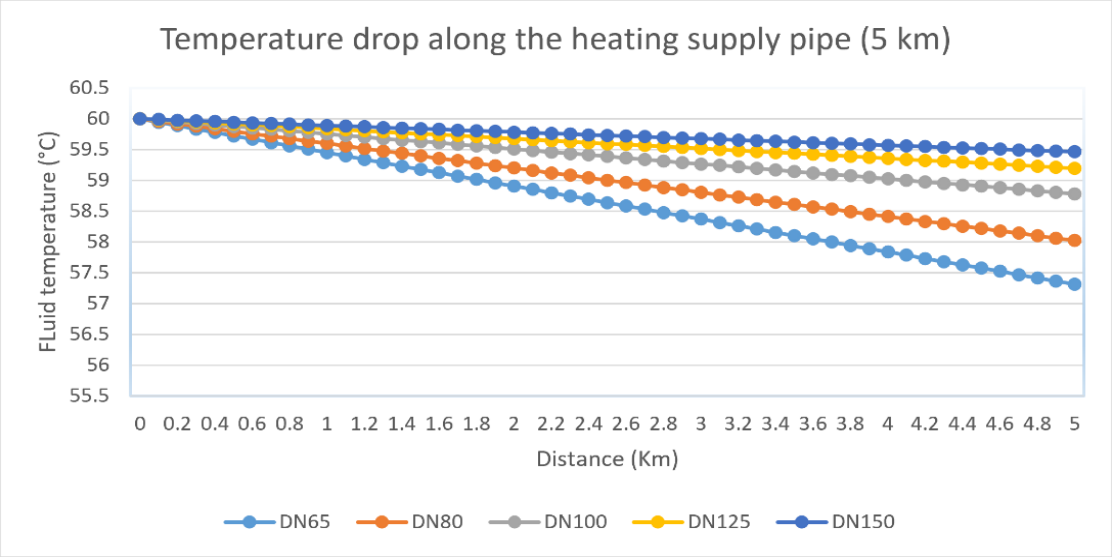


Figure 4:Temperature drop along the Heating supply pipe.

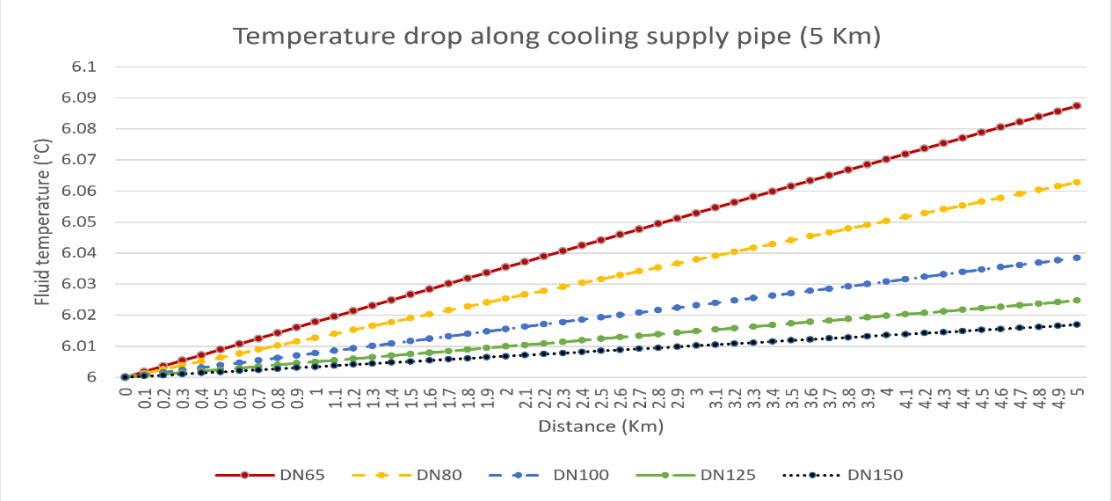


Figure 5:Temperature drop along the cooling supply pipe.

The cooling line is far less sensitive to pipe size than the heating line. Across all diameters considered, the cooling side temperature rise remains below 0.09°C over the full 5 km route. At the same time in the heating line, over the full 5 km run, the smallest pipe considered (DN65) loses just over 2°C, ending at approximately 57.8°C, while the largest (DN150) loses less than half a degree, ending at approximately 59.6°C.

5.3 Heat pump performance

The starting point for evaluating the proposed system is how well the heat pump performs. A considerably larger temperature lift than a conventional single-service heat pump normally must manage. Since it is effectively serving both ends of a district energy system advance.

5.3.1 Ideal balanced case

The reference case assumes heating and cooling demand are equal, at 230 kW each. Given the supply and return temperatures on both sides (60°C/25°C on the heating side, 6°C/16°C on the cooling side). The mass flow rate required is found using equation 4.21, and combining the two streams gives a mixed return temperature of 18.6°C. At this balanced condition, a simple Carnot coefficient of performance (COP) reference to the hot supply temperature and the mixed

return temperature works out to about 8.0. This is an idealist figure useful mainly as a starting reference point rather than a realistic performance estimate.

5.3.2 Two off-design scenarios

In practice, heating and cooling demands will rarely be equal at every moment, so two deviation scenarios were examined.

1. Heating fixed at 100% (230 kW), cooling load reduced from 100% down to 10%. As the cooling distribution varies, the return stream is increasingly dominated by the hot water return. So, the mixed temperature rises from 18.1°C up to 22.7°C at 10% cooling load.
2. Cooling fixed at 100% (230 kW), heating load reduced from 100% down to 10%. Here, the opposite happens, with less hot return diluting the mix. The mix return temperature falls from 18.1°C down to 16.3°C at 10% heating load.

5.3.3 Coefficient of performance (COP) using the Carnot and the Lorenz Cycle.

Carnot coefficient of performance (COP) with a 0.95 heat exchange effectiveness and a second law efficiency of 0.45 barely moves at across both scenarios – from 2.654 down to 2.652 in the heating fixed case and a constant 2.65in the cooling fixed case. The two scenarios move slightly in opposite directions, but the total spread across the entire 10% to 100% load range in either direction is under 0.003, which is negligible for practical purposes.

The Lorenz cycle coefficient of performance (COP) calculated using the log mean temperatures for the condenser and evaporator glides is completely insensitive in both load scenarios. It holds coefficient of performance (COP) values of 3.0, 3.4 and 4.0 for assumed second law efficiencies of 0.45, 0.50 and 0.60, respectively. At every single load point in both heating fixed and cooling fixed cases, this follows directly from how the calculation is carried out. For each load point, the refrigerant saturation temperature at the evaporator and condenser is found from the fixed network supply temperature together with the load-dependent mixed return temperature using formula 4.20. Heat exchanger effectiveness is taken as 1.0 because the second law efficiency range applied (0.45 to 0.60) is itself reported by (Lund, et al., 2017) as already incorporating heat exchanger losses. Therefore, the load-dependent return temperature cancels out, leaving the refrigerant saturation temperature equals the fixed supplied temperature. The 5 K superheat and subcooling margins are applied on top of these saturation points before the log-mean temperatures are calculated.

Taken together, these results show that the more physically grounded Lorenz and effectiveness-based Carnot methods both show that this coefficient of performance (COP) itself is remarkably stable.

5.3.4 Thermal storage requirement

Since the heat pump is sized to serve a fixed 230 kW duty on one side while the other side demands fluctuate with load. The two flow streams no longer balance once the loads deviate from the ideal case. The excess flow from whichever side remains at full output cannot simply be ignored. it must be diverted to storage rather than sent back to the heat pump. Since the heat pump itself only needs enough return flow to match its fixed duty.

5.3.5 System Energy Balance

The system energy balance shows how the heating and cooling demands relate to the external heat input. The compressor electricity follows from the heating demand and the Coefficient of

Performance (COP), and the COP does not change with load. So a change in cooling demand does not change the electricity; it changes the heat available at the evaporator. When cooling demand is low compared with heating demand, the cooling return supplies less heat to the evaporator, and the auxiliary heat from the data centre must make up the difference. When cooling demand is high compared with heating demand, the heat pump rejects less heat than the system receives, and the surplus must be rejected, for example, by a chiller. The balance is therefore easiest to maintain when heating and cooling demand are of similar size.

5.4 Cost comparison

The final piece of analysis is the comparison of installation cost of three configurations considered throughout the thesis. The proposed three-pipe system, a conventional four-pipe system representing two fully independent heating and cooling networks and the Ranagård system. Considering category, a (inner city) factor of 1.3 to reflect a dense urban installation route, the cost becomes:

Table 8: Cost comparison.

Category	Three-pipe system (sek/m)	Four-pipe system (sek/m)	Ranagård (sek/m)
Pipe Material + pipework + casing	8444.8	13821.6	6912.1
Land and construction work	16313.7	26704.6	13351
Design and inspection	3805.1	6164.6	3082.3
Surveying/permits / miscellaneous	679.9	1112.8	556.4
Internal labour time	1110.2	1817.4	908.7
Total Cost	30353.7	49621	24810.5

The three-pipe system comes in at roughly 39% lower total cost per meter than the four-pipe equivalent alternative. A saving of nearly 19300sek per meter off route. This is the clearest argument in favour of the shared return three pipe concept. Against the Ranagård system, the three-pipe system is about 22% more expensive per meter. This should not be read as a disadvantage in the same way as the four pipe comparison. The Ranagård reference delivers heating only, but this three-pipe system can provide heating and cooling simultaneously from the same trench. This system offers a well-supported capital cost advantage over running independent heating and cooling networks.

6 Discussion and Conclusion

The final chapter synthesises the findings of the thesis, interprets them against the research questions presented in Chapter 1, and sets out the principal conclusions of the study. It responds to each research question, discusses the broader implications of the findings for district energy policy, urban planning and engineering practices. This chapter also addresses the scope for future research to build on and extend the work. The chapter concludes with a statement of the thesis's principal contributions to knowledge and closing conclusions.

The study was motivated by a clearly identified gap in the literature. Three-pipe district heating and cooling configurations combining two supply pipes with a single shared return have been analysed in prior works by (Arabkoohsara, et al., 2019) (Xu, et al., 2021) (Xu, et al., 2026), but no prior study had examined a unidirectional three-pipe configuration combining a heating supply (Pipe 1 = 60°C), a shared common return pipe (Pipe 2 = 18-23°C), and a cooling supply pipe (Pipe 3 = 6°C), in which a centralised dual evaporator heat pump recovers heat between simultaneous heating and cooling demand through a Y-separator, and is further supported by a secondary, load-independent waste heat source from a data centre at approximately 25°C. This thesis has addressed that gap through a structured analytical methodology, combining a validated multi-pipe heat transfer model, a mass flow weighted mixed-return temperature model, and two independent heat pump performance methods (Carnot and Lorenz), producing a transparent framework for evaluating the proposed system and comparing it against the conventional four-pipe network and the Ranagård three-pipe installation.

Although the pressure drop study is not included in this thesis, we acknowledge the pressure drop in the mixed-return pipe. The most viable option is to increase the pipe diameter. In this thesis, however, the distribution network uses a pre-insulated twin-pipe configuration, with hot supply and shared return of DN65 housed in a single casing. Since both lines share the same nominal diameter, enlarging the return pipe would require a larger twin-pipe size for the whole section, with consequences for investment cost and heat losses. Because this work is a model study intended to demonstrate the feasibility of the proposed system rather than produce a final design, detailed pipe optimisation was considered outside its scope. It is recommended as apriority for future work, including comparison of alternatibe diamerters in terms of pressure loss, pumpng power , heat loass and cost.

6.1 Response to research questions

6.1.1 Primary research question

The primary research question asked: Can a three-pipe district heating and cooling configuration improve temperature differentials and system efficiency compared with conventional parallel networks?

The analytical framework developed in Chapter 4 supports a positive answer in general. Because the central heat pump draws on a moderate mixed return temperature in Pipe 2, rather than a cold ground source or ambient air, and a second largely load-independent heat source is available from data centre waste heat at approximately 25°C. Thus, the temperature lift required to reach the 60°C heating supply and the temperature depression required to reach the 6°C cooling supply are both smaller than for a conventional system drawing on ambient conditions. Since the heat pump coefficient of performance (COP) falls as temperature lift increases (Schlosser, et al., 2020) this reduced lift improved the effective temperature differential (ΔT) than the four-pipe system. The magnitude of those improvements across the different load scenarios is quantified in Chapter 5 by applying the Carnot and Lorenz heat pump models to the calculated Pipe 2 mixed return temperature under different load scenario.

6.1.2 Sub-Question 1: Advantages and Limitations

Sub-question 1 asked what are the potential advantages and practical limitations of the three-pipe system are?
The principal advantages identified in Chapter 3 are: First, replacing the two separate return pipes of the four-pipe standard with a single shared return pipe (Pipe 2) reduces the total pipe count from four to three, lowering material, trench excavation insulation, and labour cost. The economic comparison in Chapter 4 quantifies this using five-category installed cost model (material, civil works, design, permitting and labour) benchmarked against the (Energiföretagen, 2022) categories, applying the Category A (inner-city) factor of 1.3 appropriate to a dense urban installation route. Second, the moderate mixed return temperature in Pipe 2 provides the central heat pump with a more favourable source and sink condition than ambient air to ground. Third, the shared return, Y-separator, dual evaporator enables continuous internal heat recovery between the heating and cooling circuits without requiring dedicated heat recovery equipment, an advantage four-pipe system does not offer. Fourth, the dual evaporator configuration allows the plant to draw on a second heat source from data centre waste heat, improving average coefficient of performance (COP) and output availability during periods when the district-side evaporator load is reduced by the demand.
The principal limitations also identified in Chapter 3 , are the loss of independent control over the heating and cooling return temperatures, since the mixed return temperature has the property of the load conditions rather than a directly set value. Increased hydraulic complexity at each substation connection point and at the Y-separator, where pressure balancing is required to prevent reverse flow between heating and cooling circuits. The risk of inter-pipe heat transfer between Pipe 1 and Pipe 3, which are separated by a temperature differential of approximately 54°C. And the absence of any commercial track record for this specific heating and cooling configuration creates procurement risks in market.

6.1.3 Sub-Question 2: Return temperature and ΔT

Sub-question 2 asked how the shared return pipe affects the mixed return temperature and the system wide temperature differential under different load scenarios?
The mixed return temperature model in Chapter 4 defines T mix in Pipe 2 as the mass flow weighted average of the heating circuit return and the cooling circuit return, so that the T mi moves toward the upper end of the design range (approximately 23°C) under peak winter, heating dominant conditions and toward the lower end (approximately 18°C) under peak summer, cooling dominant conditions, with an intermediate value expected under the simultaneous load scenario. In every scenario, T_{mix} sits in a moderate range that is thermodynamically more suitable for the central heat pump. Because the central heat pump draws on the T mix as the source, the effective temperature lift to the 60°C hot supply and the effective temperature depression to the 6°C cold supply is both depends on where T mix sits within 18-23°C range.

6.1.4 Sub-Question 3: Heat recovery potential

Sub-question 3 asked what the theoretical potential for heat recovery is between heating and cooling consumers through the shared return loop?
The plant architecture set out in Chapter 3, a Y-separator dividing the Pipe 2 mixed return stream between the evaporator and condenser sides of a single central heat pump- means that heat recovery between the cooling and heating circuits is continuous operation rather than an optional add-on. Every unit of heat extracted from the evaporator sub stream to produce 6°C cold supply is together with the compressor work input, delivered to the condenser sub stream to produce the 60°C hot supply. Only primary energy input to this system is the electrical

energy consumed by the compressor. The degree of recovery achieved in practice depends on the instantaneous split ratio set by Y-separator, which in turn depends on the relative magnitude of heating and cooling demand. The qualitative finding is that the internal heat recovery mechanism can reduce or eliminate the need to reject the condenser heat to the environment, an advantage the four-pipe system do not offer.

6.1.5 Sub-Question 4: Pipe heat transfer influence

Sub-question 4 asked how pipe heat transfer influences system performance?
The pipe heat transfer model in Chapter 4 accounts for three effects acting on each of the three buried pipes: conduction to the undisturbed soil through the combined pipe-wall and soil resistance, mutual interference with each of the other two pipes via resistances R12(Pipe 1-Pipe 2), R13 (Pipe 1 – Pipe 3), and R23 (Pipe 2- Pipe 3), calculated from the centre-to-centre spacing of each pair and specifically for Pipe 1- Pipe 3 pair, a direct hot-to-cold heat flux that determines the temperature rise of the cooling supply along the pipe route. Because Pipe 1 (60°C) and Pipe 3 (6°C) have the largest temperature differential of any air in the system, at approximately 54°C, the design response adopted in Chapter 3 was to place Pipe 2 together with Pipe 2 inside a single shared twin casing at a close spacing, while installing Pipe 3 in an independent casing offset by a 150 mm trench gap, giving twin casing to cold casing centre distance of 207 mm. What can be concluded qualitatively is that pipe heat transfer in this system is a manageable design variable, governed by casing geometry and insulation specification.

6.1.6 Sub-Question 5: System comparison

Sub-question 5 asked how the proposed three-pipe configuration compares in efficiency, cost, and technical complexity with the conventional four-pipe parallel network, the bidirectional Ranagård three-pipe installations?
Against the conventional four-pipe system, the proposed configuration is expected to show an advantage on cost, since it eliminates one of the four distribution pipes, and on internal heat recovery, since the four-pipe systems fully separate heating and cooling circuits offer no equivalent mechanism. Against Ranagård installation in Sweden, one of the few documented real-world three-pipe networks, the comparison is structural rather than direct, Ranagård which uses one supply pipe carrying hot water and two return pipes and the installation is used in this thesis only cost benchmark to demonstrate that a shared-return topology can achieve measurable capital savings and stable operation in practice. Since Ranagård does not combine heating and cooling on a shared return and therefore no equivalent for the heat recovery mechanism central to the system proposed here.

6.2 Broader implications of the findings

6.2.1 Implications for district energy engineering practices

From an engineering practice perspective, the principal implication of this thesis is that the three-pipe, dual evaporator topology deserves a serious consideration as a standard design option for new mixed use district energy projects with an available low-grade waste heat source. The analytical framework developed in Chapter 4 the pipe heat transfer model, the mixed-return temperature model, and the two independent heat pump performance methods, provides a transparent, reproductible design tool that practitioners can apply with site-specific load data, soil properties, and waste heat availability. The specific pipe alignment and casing parameters adopted in Chapter 3, including the split casing arrangement, provides a concrete starting point for a detailed design.

6.3 Recommendations for future research

The findings and limitations of this thesis point to an agenda for future research, structured across three stages of increasing technical and practical background.

6.3.1 Enhanced analytical and simulation studies

The immediate next step is a simulation model capable of transient load variations over a full annual operating cycle, using TRYNYS or a modelica based framework, driven by hour-to-hour weather and occupancy based building load profiles for a representative mixed use district, and including a realistic control strategy for the Y-separator split ratio and the dual-evaporator that responds to real time mixed return temperature and also with the availability of data centre waste heat. Alongside this, a computational fluid dynamic (CFD) study of the inter-pipe heat transfer in the specific split casing geometry adopted in Chapter 3, parallel to the CFD study of (Arabkoohsara, et al., 2019) would give a more quantification of Pipe 1 – Pipe 3 heat transfer. And using this specifically test the adopted 207 mm twin-to-cold casing spacing combined with series 3 insulation in Pipe 3 is sufficient to hold cooling-supply degradation within acceptable tolerance over the 5000 m segment length.

6.3.2 Pilot installation and experimental validation

The most important step is developing the concept is the construction and monitoring of a small-scale pilot installation, sized to a district of five to twenty buildings with combined peak heating and cooling capacity in the range assumed in this thesis (230 KW). But to limit financial risk while testing the system's thermal and hydraulic behaviour under real load conditions, a primary network segment shorter than 5000 m can be used. The pilot should be instrumented to measure supply and return temperatures at the central plant for all three pipes, the mixed return temperature at multiple points in Pipe 2, mass flow rates in each circuit, the Y-Separator split ratio, the contributions of district side and data centre side evaporators to total heat pump output, and the central plant electrical energy consumption. All these must monitor over a minimum of two full annual cycles to capture both heat dominant and cooling dominant seasonal conditions.

6.3.3 Control algorithm development and optimisation

The variable mixed return temperature in Pipe 2, combined with the dual evaporator plant's two available heat sources, creates a need for adaptive control strategy that can optimise the Y-separator split ratio and the district side/data centre side evaporator balance across the full range of operating conditions. Future research should develop and test a model predictive control (MPC) strategy that uses forecast building load profiles, weather predictions, and data centre heat availability to pre-position the heat pump operating point ahead of anticipated change in mixed return temperature, also fault detection routines capable of identifying substation return temperature anomalies that could destabilise T mix.

Table 9:Recommendation for future research agenda.

Research priority	Type	Recommended tool/ method	Time horizon	Key output
CFD inter-pipe heat transfer study	Computational	ANSYS Fluent	1 year	Refined cooling degradation model
Hydraulic network model	Analytical	EPANET-DH/TERMIS	1 year	Storage tank sizing
Techno-economic lifecycle analysis	Economic	Cost model analysis	1 year	NPV, payback period

Small-scale pilot installation	Experimental	Instrumented district	3-5 years	Validated performance data
Model predictive algorithm	Control	Python/MATLAB	2-3 years	Demand response capability
Dual-source heat pump design study	Engineering	Cycle simulation	1-2 years	Combining relation of both evaporators
Regulatory and standards framework	Policy	Stakeholder engagement	2-3 years	Standardised design guidelines for three-pipe 4GDHC

6.4 Contributions to knowledge

This thesis makes the following original contributions to the body of knowledge on fourth-generation district heating and cooling systems and multi-pipe distribution topologies.
Definition of a novel unidirectional three-pipe 4GDHC topology combining a heating supply pipe (Pipe 1 =60°C), a shared common return pipe (Pipe 2), and a cooling supply pipe (Pipe 3 = 6°C), served by a centralised dual-evaporator heat pump with a Y-separator and a secondary load independent waste heat source. Previous three pipe studies (Arabkoohsara, et al., 2019) (Xu, et al., 2026) (Xu, et al., 2021) (Averfalk & Werner, 2018)and the Ranagård installation addresses two heating supply temperatures shares a return, the heating and cooling, dual source configuration examined here is a novel contribution.
A mass flow weighted mixed return temperature model that makes explicit how T mix in Pipe 2, and the consequent heat pump source condition depends on the instantaneous heating and cooling load, together with the framework for expressing both heating and cooling coefficient of performance (COP) as function of this single shared source temperature.
A structural specification, using Y-separator and dual evaporator plant architecture under which continuous heat recovery between heating and cooling consumers is achieved without dedicated heat recovery equipment. And how a secondary data centre waste heat source can be combined with the district side evaporator to improve the average coefficient of performance (COP) and output availability.
A structures multi-criteria comparison framework against three references: the conventional four-pipe network, and the Ranagård three-pipe installation, also combining five-category installed cost model benchmarked against Swedish district heating cost data with thermodynamic performance model, providing a reference framework or practitioners evaluating district energy technology for mixed use urban districts.
A transparent, reproducible analytical framework and design tool, the complete set of governing equations in Chapter 4, together with the design parameters and assumption in Chapter 3, that practitioners can apply directly with site specific data.

6.5 Conclusion

This thesis has investigated the performance potential of novel three-pipe fourth-generation district heating and cooling system, in which a heating supply pipe at 60°C and a cooling supply pipe at 6°C share a common return pipe that receives mixed return flows from both heating and cooling consumers, and in which a centralised dual evaporator heat pump fed by this shared return and by a secondary waste heat recovers from a data centre. Through a structured analytical methodology built on a validated multi-pipe heat transfer model, a mass flow weighted mixed return temperature model, and two independent heat pump performance methods, and through a comparative framework set against the conventional four-pipe network, and the Ranagård installation, this thesis establishes the theoretical basis on which the proposed configuration is expected to offer thermodynamic and capital cost advantages over the conventional system.

The central finding of the thesis is that the shared return pipe which is the defining structural feature of the proposed system is not merely a cost saving simplification but a thermodynamic asset. By maintaining a moderate mixed return temperature between the extremes of the heating and cooling return streams, Pipe 2 provides the central heat pump with a more favourable source and sink condition than a conventional ground source or ambient air. And the Y-separator and dual evaporator plant architecture convert this favourable condition into a structural, continuous heat recovery mechanism between heating and cooling consumers that the four-pipe system achieves.

The principal limitations of the study, the steady-state modelling assumption the use of assumed rather than a measured input data, the simplified treatment of dual source heat pump performance and absence of field validation are acknowledged and motivate a structured agenda for future research.

The proposed system is positioned most advantageously in mixed-use urban districts with significant simultaneous heating and cooling demand, access to a low-grade waste heat source such as a data centre, and a modern low-energy building with a 60°C heating supply. In this context, the three-pipe, dual-source configuration offers a technically credible and thermodynamically distinctive contribution to the portfolio of urban energy solutions and provides analytical tools to adopt inter-pipe casing spacing and combining two evaporator sources, which are needed to carry the concept from theoretical framework to pilot-scale demonstration.

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